

Eslam Shokr

Doctoral Researcher in Experimental Particle Physics

eslam.shokr@cern.ch [LinkedIn](#) [GitHub](#)

Research contributions

Tau precision physics

Subleading analyzer: $\gamma\gamma \rightarrow \tau\tau$ production and tau anomalous magnetic moment in ultraperipheral Pb-Pb collisions. Approved as CMS PAS HIN-24-011.

BSM resonance searches

Lead analyzer: model-agnostic $X \rightarrow YY' \rightarrow bbWW$ search with CMS Run 2 and Run 3 data, using a three-dimensional likelihood fit.

Phase-II detector R&D

Tracker Endcap Pixel R&D for High-Luminosity CMS: eight-module calibration framework, detector testing and silicon-sensor characterization.

Key skills

Programming

Python, C++, Bash, CMake, Linux

Analysis & simulation

ROOT, CMSSW, Combine, PYTHIA, HERWIG, UPCgen, STARlight, Geant4

Languages

Arabic (native), English, German (A1)

Relevant experience

Sep 2022 — Present

Doctoral Researcher

University of Zurich · Zurich, Switzerland

Teaching assistant for undergraduate physics and laboratory courses.

Jun 2021 — Present

Assistant Lecturer · on study leave

Mansoura University · Egypt

Academic appointment in the Physics Department, Faculty of Science.

Feb — Jul 2022

Erasmus+ Research Fellowship

Sapienza University of Rome · Italy

CMS research training; coursework in particle physics, detectors, accelerators and machine learning.

Dec 2016 — May 2021

Teaching Assistant

Mansoura University · Egypt

Taught undergraduate laboratories in properties of matter, electromagnetism, electronics and optics.

Education & scholarship

Sep 2022 — Present

PhD, Experimental Particle Physics · ongoing

University of Zurich · Switzerland

Supervisor: Prof. Ben Kilminster. Swiss Government Excellence Scholarship (2022–2025).

May 2018 — Apr 2021

MSc, Particle Physics

Mansoura University · Egypt

Thesis: charged-particle production in CMS pp collisions. Supervisors: Prof. Ali El-Farrash, Prof. Albert De Roeck, and Dr. Mohammed Attia Mahmoud.

Sep 2012 — Jul 2016

BSc, Physics

Mansoura University · Egypt

Graduated first in the program with excellent honors (85.2%).

Supervision & mentoring

Jul — Sep 2026

Summer student supervisor

Muon entanglement in $Z \rightarrow \mu^+\mu^-$ events

Identified a theoretical proposal and initiated its experimental application at CMS. Supervised a summer student on $Z \rightarrow \mu^+\mu^-$ reconstruction and entanglement using kinematic spin correlations.

The student presented the project in Bern and received a best-talk award.

[View student project presentation \(PDF\)](#) ↗

Publications & results

Eslam Shokr (CMS Collaboration) · Measurement of anomalous magnetic moment of tau lepton at CMS. Technical report. [CERN Document Server](#)

CMS PAS HIN-24-011 · Measurement of $\gamma\gamma \rightarrow \tau\tau$ production and constraints on the tau anomalous magnetic moment in ultraperipheral Pb-Pb collisions. [CMS analysis note \(PDF\)](#)

E. Shokr, A. De Roeck & M. A. Mahmoud · Modeling of charged-particle multiplicity and transverse-momentum distributions in pp collisions using a DNN. Scientific Reports, 2022. [Read article](#)

E. Shokr, A. H. El-Farrash, A. De Roeck & M. A. Mahmoud · Study of charged-particle multiplicity fluctuations in pp collisions with Monte Carlo event generators at the LHC. IJMPE, 2020. [Publisher](#) · [DOI](#)

Selected talks & posters

27 Aug 2026

Model-agnostic $X \rightarrow YY' \rightarrow bbWW$ search with CMS

Talk · Swiss Physical Society Annual Meeting 2026 · EPFL, Lausanne

Oct 2025

Limits on the tau anomalous magnetic moment in CMS

Plenary talk · TAU 2025 · Marseille

18 Aug 2025

Exclusive dimuon photoproduction in ultraperipheral Pb-Pb collisions

Joint Annual Meeting of ÖPG and SPS · Vienna

6 Apr 2025

Measurement of the tau $g-2$ factor in ultraperipheral Pb-Pb collisions

Poster · Quark Matter 2025 · Frankfurt

18 Nov 2024

Measurement of the tau $g-2$ factor with CMS

LHC students poster session · LHCC · CERN

Academic service

2022–2026

Weekly research group meeting coordination

Coordinated weekly group meetings for four years: scheduling, room bookings with departmental administration, and technical support for hybrid presentations.

References

Prof. Dr. Ben Kilminster

Group Leader, Zurich CMS group
University of Zurich · Switzerland

Prof. Dr. Albert De Roeck

Senior Physicist
CERN · Switzerland

Asst. Prof. Mohammed Attia Mahmoud

Group Leader & Director, CHEP-FU ·
Fayoum University, Egypt